DATA ITEM DESCRIPTION

Form Approved
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Public reporting burden for this collection of information is estimated to average 110 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503.

1. TITLE	2. IDENTIFICATION NUMBER
AIRCRAFT ELECTRICAL COMPATIBILITY DATA (ECD)	DI-NUOR-81410

3. DESCRIPTION/PURPOSE

3.1 The ECD defines the electrical power including the physical and the logical signal interfaces through the avionics hardware components to the nuclear weapons for all weapon configurations. The ECD covers all of the electrical and electronic interfaces in the monitor, control and delivery of nuclear weapons, whether those interfaces are discrete lines, or multiplex data busses.

(Continued on Page 2)

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
940729	F/SA-ALC-NWI		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for data resulting from the work task described by 90.5 of MIL-STD-1822.

7.2 This DID is related to DI-NUOR-81409 , Nuclear Certification Plan, DI-NUOR-81408 , Aircraft Monitor and Control Design Analysis Report, and DI-NUOR-81405 , Aircraft Nuclear Safety Analysis Report.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		F7047

10. PREPARATION INSTRUCTIONS

10.1 Format. Contractor format is acceptable.

10.2 Content. The ECD shall define the interfaces from the operator's input devices to the connector that physically connects to the nuclear weapon.

10.2.1 The ECD shall provide engineering data and descriptions that define:

a. The general system information to include ECD introduction and index; abbreviations acronyms; aircraft monitor and control and delivery system block diagrams; system definition and overview (short text); interface table describing weapons, training devices and test sets; nuclear configuration table identifying hardware and its location; and panel arrangement.

b. The avionics system logical signal interfaces throughout the avionics system components as they relate to monitoring and controlling of logic signals for nuclear weapons and their delivery systems.

c. The physical interfaces showing wire identification, wire gages, wire lengths, connector pin identification, bulkhead and splice locations.

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11. DISTRIBUTION STATEMENT

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Block 3, Description/Purpose (Continued)

The ECD documents the digital and analog electronic design used to monitor, control, and deliver a nuclear weapon.

Block 10, Preparation Instructions (Continued)

d. The avionics system electrical interfaces for controlling the application of electrical power to the nuclear weapons and controller line replaceable units, and the monitor returns throughout the avionics system which determine weapon and delivery system status.

10.2.2 The ECD shall provide:

a. Provisions for recording the coordination and approval by cognizant government agencies and their contractors.

b. References to applicable DOE (Sandia National Labaratory) electrical compatibility control drawing(s).

c. A listing of electrical test equipment to be used with the aircraft monitor and control system.

d. The sequence of logical and control events of the aircraft's nuclear weapon delivery system from takeoff to the launch or release of a nuclear weapon.

e. The characteristics of the aircraft's electrical power generation and power distribution system in relation to the appropriate military standards.

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